

An Identifiability Audit of Evolutionary Oncology

What standard-of-care measurement can and cannot recover —
and what it would take

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Note on structure. **Part I** (§§1–10) states what we believe. **Part II** (§11) states how we got there — including every hypothesis of our own that we had to abandon, and the one a referee had to find for us.

Abstract

Adaptive and evolution-guided cancer therapies rest on quantitative claims about tumour population dynamics. We ask a prior question: **given the data oncology actually collects, can those claims be tested at all?**

They cannot. Fitting a two-population competition model to simulated patients under uninterrupted maximum-tolerated-dose (MTD) therapy with q8w restaging, **roughly half (50–57%) of honest maximum-likelihood fits conclude that resistance is free or advantageous when its true fitness cost is 30%** — worse than a coin flip, with a median estimate of $\hat{c} = -2.17$. The competition coefficient is worse still: its estimate wanders the entire admissible range. The Fisher information matrix has condition number $\sim 10^{10}$ and the likelihood is flat to the boundary. These parameters are not imprecise; they are *unidentified*.

Denser sampling does not fix this. A sevenfold increase in blood draws leaves the sign-error rate exactly unchanged. Identifiability requires a *conjunction* of three things — a clone-resolved observable, a window with the drug **off**, and sampling concentrated around that window — and removing any one collapses it. Because treatment interruptions already occur in routine care for clinical reasons, the triad can be assembled **without prescribing any change in treatment**. Under this design the fitness cost resolves at $n \approx 7$ patients and the directional suppression coefficient at $n \approx 10$, at ~ 18 blood draws each.

Crucially, **not every interruption is usable**. A hold for Grade 3/4 toxicity is not a clean perturbation but a systemic crisis, and if the host insult spares the resistant clone — biologically plausible — it biases the fitness cost *downward*, reproducing precisely the error the design exists to prevent (sign-error returns to 58%). The identifying set must be restricted to interruptions that perturb the drug and nothing else: insurance lapses, supply interruptions, scheduled surgery, non-adherence. The design can test this assumption on itself.

We then ask what the data *should* be asked for, and find the field is asking for the wrong thing. **The benefit of adaptive therapy is driven by competition, not by the fitness cost of resistance.** A fitness cost is neither necessary (36 days of benefit when resistance is *free or advantageous*) nor sufficient (4 days when the cost is large but competition is weak). Competition carries roughly twice the leverage. This replicates in a mechanistic consumer–resource model where competition *emerges* from resource sharing rather than being a free

coefficient, and where a genuine zero-competition control exists (leverage ratio 1.65 vs 1.95).

Finally, **scheduling requires no model at all**: closed-loop feedback captures 86% of an oracle's gain, where an open-loop schedule from a misspecified model captures 6%.

PART I — WHAT WE BELIEVE

1 Introduction: a field with structural, not conceptual, deficits

Cancer research does not suffer from a shortage of ideas or of rigour. It suffers, like any large scientific enterprise, from deficits that are *structural* — ones that no individual investigator is incentivised to repair, however clearly they see them. We list them not as an indictment but to locate this paper among them, and because the last item is the one we can actually do something about.

- (i) **Cancer is an evolving population, treated as a static target.** Resistant subclones are near-inevitable in large heterogeneous metastatic disease [1], and MTD therapy plausibly *accelerates* their expansion by removing the sensitive competitors suppressing them — competitive release. Adaptive therapy has been formalised for over a decade [2], validated preclinically [3] and tested in metastatic castrate-resistant prostate cancer [4]; MTD remains standard of care.
- (ii) **Metastasis and dormancy — what actually kills patients — are studied less than primary tumour biology,** which is more tractable and more publishable.
- (iii) **Much of the foundational preclinical literature does not replicate** [5–8], and no funded institution exists whose job is to confirm a preclinical result before it consumes a phase III.
- (iv) **Prevention and causation are underweighted relative to treatment,** despite prevention accounting for several of the largest historical wins.
- (v) **Overdiagnosis and surrogate endpoints corrupt the measurement of success.** Progression-free survival is not survival.
- (vi) **Nobody runs the trials nobody profits from** — off-patent agents, generic combinations, de-escalation, schedule variation.
- (vii) **The measurement protocol cannot identify the quantities the models depend on.**

This paper is about (vii). It is **not** a claim that adaptive therapy works. It is a narrower and more defensible claim: **under current measurement practice, adaptive therapy's central premises are not testable** — and this is fixable, cheaply, without asking any clinician to treat any patient differently.

2 The two candidate parameters

Adaptive therapy's mechanism is *competitive suppression*: keep drug-sensitive cells alive so they hold resistant cells in check. Two parameters could underwrite this, and the field has fixated on one of them.

The fitness cost of resistance, c . Resistant cells are assumed to grow more slowly without drug: $r_R = r_S(1 - c)$, a life-history trade-off [9] with direct empirical support in EGFR-mutant lung cancer [10]. The standard framing is that adaptive therapy works *if and only if* $c > 0$ [2]. This is the parameter the literature argues about.

The competition coefficient, α . Specifically α_{SR} : the degree to which sensitive biomass denies carrying capacity to resistant cells. There is direct experimental evidence that spatial competition alone constrains resistance [11]. This is the parameter nobody measures.

We will show that neither is identifiable under standard of care; that both become identifiable under a cheap observational redesign; and that α , **not c , is the one that decides the question.**

3 Model and audit design

Dynamics. Two competing populations under therapy:

$$\frac{dS}{dt} = r_S S (1 - (S + \alpha_{RS} R)) - d u(t) S \quad (1)$$

$$\frac{dR}{dt} = r_R R (1 - (R + \alpha_{SR} S)), \quad r_R = r_S(1 - c) \quad (2)$$

with $u(t) \in \{0, 1\}$ the drug indicator. Truth: $r_S = 0.03/\text{d}$, $c = 0.30$, $\alpha = 1.0$, $d = 0.05/\text{d}$, initial resistant fraction $f_0 = 0.01$.

Observation. ctDNA is cleared from plasma with a half-life of minutes to a few hours [12–14], so plasma concentration is a near-instantaneous readout of current shedding rather than a smoothed average. Shedding has a burden-proportional and a death-proportional component, the latter reproducing the transient ctDNA *rise* seen days after cytotoxic initiation [14]:

$$y(t) = (S + \rho R) + \kappa d u(t) S \quad (3)$$

Two observables are compared:

- **total** — plasma ctDNA burden alone (a tumour-agnostic assay), 30% CV lognormal noise.
- **clone-resolved** — burden **and** the resistant allele fraction (a tumour-informed panel), on the logit scale.

Protocols. *Continuous* MTD (standard of care); and an *incidental hold* — an unplanned interruption of L days, of the kind that occurs routinely for toxicity, dose reduction, surgery, insurance gaps, or non-adherence.

Inference. We distinguish *structural* from *practical* non-identifiability throughout [15, 16], following the approach of Eisenberg and Jain [17] for cancer chemotherapy models. We use Fisher information and Cramér–Rao bounds, validated throughout against **multi-start** maximum likelihood on synthetic patients, with c free on $[-3, 0.99]$ so a fit is *permitted* to conclude resistance is beneficial. Restarts are drawn from the whole parameter box, not perturbed from the truth.

The test. Not “is c estimated precisely” but the weakest possible bar: **can we determine its sign?**

4 Under standard of care, nothing is identifiable

4.1 The fitness cost

Sampling	Draws	cond(FIM)	SE(c)	Sign resolved?
standard (q8w)	6	6.6×10^{10}	5.21	no
q4w	11	5.8×10^{10}	3.72	no
q2w	21	6.3×10^{10}	3.03	no
weekly	41	6.8×10^{10}	2.44	no
daily	281	7.6×10^{10}	1.07	no

Total ctDNA, continuous MTD. Daily sampling for forty weeks — 281 draws — fails. The condition number is $\sim 10^{10}$ throughout and barely moves with cadence: adding measurements does not add information about c .

The estimator correlation matrix shows why: $\hat{c}$ correlates with $\hat{r}_S$ at 0.996, and $\hat{\alpha}$ with $\hat{f}_0$ at 1.000. **Total burden cannot distinguish a slow-growing resistant clone from a fast-growing clone that started rarer.** The clonal split is not in the observable.

At these condition numbers a Cramér–Rao bound is a linearisation of a near-singular surface, so we fit synthetic patients directly. This degradation with parameter count is consistent with what is already known for simpler growth models: two-parameter models are practically identifiable while three-parameter models frequently are not, even on dense preclinical data [18, 19], and lack of practical identifiability is a known route to confidently wrong predictions [20].

4.2 Multi-start MLE: worse than the bound suggests

Design	median $\hat{c}$	95% interval	fits concluding resistance is FREE or BENEFICIAL
status quo — total, q8w, MTD	−2.17	[−3.00, +0.88]	50–57%
dense only — total, weekly , MTD	−0.34	[−3.00, +0.86]	57%
clones only — cloneAF, q8w, MTD	+0.25	[−3.00, +0.74]	37%
proposed — cloneAF, burst, 28d hold	+0.33	[−0.06, +0.51]	7–10%

The lower bound hits the *box constraint* in the first three rows: the likelihood is flat to the boundary. The parameter is unidentified, and the “SE” of 5.21 is a meaningless number attached to a flat surface.

In a world where resistance carries a substantial 30% fitness cost, a competent analyst fitting a competent model to competently collected data concludes that **resistance is free or advantageous roughly half the time**. The median honest estimate is $\hat{c} = -2.17$: confidently, wildly, in the wrong direction.

4.3 The competition coefficient is worse

Design	median $\hat{\alpha}$	95% interval	$\alpha = 0$ excluded?
status quo	3.000 (<i>pinned at bound</i>)	[0.000, 3.000] — the entire box	no
proposed	0.867	[0.471, 1.573]	yes

Under standard of care the estimate for α wanders the whole admissible range. There is essentially no information in the data about it.

5 The identifiability triad

The three ingredients are conjunctive, not additive. Removing any one collapses identifiability.

- **Sampling density alone:** 6 draws $\rightarrow$ 41 draws (a $7\times$ increase in phlebotomy). Sign-error: 57% $\rightarrow$ **57%**. *Exactly unchanged.*
- **Clone resolution alone:** total ctDNA $\rightarrow$ total + resistant allele fraction, at q8w. Sign-error: 57% $\rightarrow$ **37%**. Better; still catastrophic.
- **Both, plus a 28-day drug-off window:** **7–10%**.

5.1 Why the drug-off window is indispensable

Under continuous MTD the sensitive population is driven toward extinction, the competition term αS vanishes, and the resistant clone expands essentially unopposed at r_R — which is confounded with r_S and f_0 .

A drug-off window is the only thing that lets the sensitive population re-express its growth advantage — which is what competitive suppression *is*. This is a system-identification truism: **you cannot identify a system you never excite**. Evolutionary oncology has been trying to.

5.2 What the interruption must look like

Hold length	SE(c)	Sign resolved (single patient)?
7 d	0.394	no
21 d	0.191	no
28 d	0.151	yes
42 d	0.107	yes

Hold *timing* matters far less than hold *length* (SE 0.177–0.247 for holds starting anywhere between days 28 and 196). This is fortunate: one cannot choose when a toxicity hold occurs, but one can choose to sample around it.

5.3 Design efficiency, in patients

Patients required to determine the sign of a subtle cost ($c = 0.15$), ideal-pooling bound:

Observable	Sampling	Protocol	Draws/pt	Patients
total	q8w	uninterrupted	6	89,424[†]
total	burst	uninterrupted	18	23,087 [†]
total	burst	28d hold	18	150
cloneAF	q8w	uninterrupted	6	102
cloneAF	q8w	28d hold	6	23
cloneAF	burst	28d hold	18	7

[†] Read as “unidentified”, not as a literal sample size — these are extrapolations from a singular surface.

The **directional** suppression coefficient α_{SR} (with α_{SR} and α_{RS} kept separate — the hard case) resolves at **n $\approx$ 10**:

patients	median $\hat{\alpha}_{SR}$	95% interval	resolved?
1	1.201	$[-1.000, 3.000]$	no
5	0.774	$[-0.733, 2.184]$	no
10	1.048	[0.472, 1.941]	yes
40	0.986	$[0.511, 1.322]$	yes

No single patient’s data will do it; a cohort of ten will. This is a *cohort* design, not a personalised one — it can establish whether competitive suppression is real in a tumour type, not whether it is operating in the patient in front of you.

6 Robustness, and the right estimand

6.1 What the design absorbs

- **Differential clone shedding.** If clones shed ctDNA at different rates ($\rho = \beta_R/\beta_S \neq 1$), the plasma allele fraction is a biased view of clonal composition. Making ρ a free parameter costs **nothing**: $SE(c)$ unchanged at 0.151. Even *wrongly* assuming $\rho = 1$ against a threefold truth biases $\hat{c}$ by only -0.037 .
- **Shedding-model (κ) misspecification.** A $2.5\times$ error biases $\hat{c}$ by -0.02 to -0.05 . Conservative in direction.
- **Between-patient heterogeneity.** Inflation is mild while $\omega < SE_1$ (n : $10 \rightarrow 14$ at $c = 0.15$ with $\omega = 0.15$).
- **Competition regime.** $SE(c) = 0.151$ for the proposed design at every α from 0.5 to 1.0. Identifiability is regime-robust.

6.2 What it does not absorb: a fixed compartment count

Real resistance is polyclonal and spatially structured [1, 21, 22]. Fitting a two-clone model to three-clone truth destroys the estimate — **and the sign of the bias is set by modelling convention, not by data**:

2-clone model variant	median $\hat{c}$	bias
rigid kill (resistant assumed fully refractory)	0.369	+0.069
+ differential shedding	0.445	+0.145
flexible kill (kill rate free)	−0.12 to −0.04	≈ −0.4, sign flips

Same data, same clone count, opposite conclusion. **Pooling divides variance by n and does nothing to bias.** At large n one converges, with tight confidence intervals, on a number whose sign was decided by a modelling choice.

6.3 The ontology is undecidable — and does not need to be decided

Within a *discrete* candidate set, clone number is recoverable **only with a per-variant panel**:

Truth	Assay	k recovered
$k = 3$	lumped allele fraction	0% — selects $k=2$ in 10/10
$k = 3$	per-variant panel	100%

The lumped assay reports a plausible-looking $\hat{c} = 0.185$ with no diagnostic complaint. Silent, confident, unfalsifiable from within the analysis.

But admit a **continuum** model — resistance as a distribution of sensitivities rather than compartments — and clone number evaporates:

Truth	$\hat{k} = 2$	$\hat{k} = 3$	continuum	correct?
CONTINUUM	1/8	0/8	7/8	yes
DISCRETE-3	2/8	0/8	6/8	no — spurious

BIC prefers the continuum in both worlds. The compartment count is not identifiable.

It does not need to be. The clone number was never the estimand. What adaptive therapy requires is a *functional* of the resistance distribution, well-defined under either ontology:

$$c_{\text{eff}} = 1 - \frac{\mathbb{E}[r(x) \mid x > 0]}{r_S} \quad (4)$$

Truth	true c_{eff}	reported $\hat{c}_{\text{eff}}$	wrong sign
CONTINUUM	0.121	0.075	0%
DISCRETE-3	0.200	0.119	0%

The sign is recovered in both worlds, under the wrong ontology, every time — with a bias that *understates* the cost, erring against the hypothesis being tested.

Two requirements are non-negotiable: a **per-variant panel** (a lumped allele fraction recovers nothing, and reports a plausible wrong answer silently), and a **continuum model in the candidate set** (discrete-only candidate sets inflate $\hat{c}_{\text{eff}}$ by 2–2.6 $\times$, in the direction that flatters adaptive therapy).

Any published fitness-cost estimate derived from a small fixed number of compartments fitted to a lumped allele fraction should be treated as **uninterpretable** — not as an estimate with wide error bars, but as a number whose sign was determined by a modelling convention.

6.4 The host is not a neutral bystander: which interruptions are actually clean

An earlier draft of this paper treated *any* incidental treatment interruption as a free excitation. That was wrong, and the error was material.

A hold for Grade 3/4 toxicity is not a clean drug-off window. The patient is in systemic crisis — hepatotoxicity, neutropenia, inflammation — and the growth and competition parameters of a tumour in a metabolically deranged host are not those of a tumour in a well one. We model this explicitly. During the hold, let $r_i \rightarrow r_i(1 + \delta_i)$ and $\alpha \rightarrow \alpha(1 + \delta_\alpha)$, generate data from the *sick* host, and fit the naive model that assumes the window is clean.

Scenario	median $\hat{c}$	bias	sign wrong
clean (non-toxic) hold	+0.301	+0.001	9%
symmetric insult (both clones $\times 0.60$)	+0.238	−0.062	29%
differential (sens $\times 0.60$, res $\times 0.80$)	+0.146	−0.154	36%
<i>same data, host effect fitted as nuisance</i>	+0.241	−0.059	22%
severe differential (sens $\times 0.40$, res $\times 0.90$)	−0.178	−0.478	58%

Three findings, and the structure of them matters more than the numbers.

(1) A *symmetric* host insult is comparatively benign. Because c is defined by a *ratio*, $r_R = r_S(1 - c)$, a factor common to both clones largely cancels. Severe symmetric slowing still degrades the estimate — not by biasing it but by *attenuating the excitation itself*, since the information in a drug-off window scales with how fast the clones diverge during it — but it does not invert the sign.

(2) A *differential* insult that spares the resistant clone is fatal. This does not cancel; it is directly confounded with c . And it is the biologically plausible case: efflux pumps, stress-response programs and quiescence plausibly confer *both* drug resistance *and* robustness to host metabolic crisis. Worse, it biases $\hat{c}$ **downward**.

The confound is aligned with the failure mode. A host insult that spares the resistant clone makes resistance *look* free — which is precisely the false conclusion this paper exists to prevent. It does not add noise; it manufactures our error, and it does so with a straight face.

(3) Fitting the host effect as a nuisance parameter does not rescue it ($36\% \rightarrow 22\%$). We tried the obvious statistical fix and it is insufficient.

The remedy is therefore not statistical but an inclusion criterion. Not every incidental hold carries a host insult. Interruptions arising from *insurance lapse or prior-authorisation delay, drug supply interruption, scheduled surgery, elective procedure, travel, non-adherence, or clinic closure* perturb the drug and nothing else. **These, and only these, are clean excitations.** Toxicity holds must be excluded from the identifying set — or, at minimum, analysed as a separate stratum and never pooled with the clean ones.

The cost is enrolment, not validity: if a fraction f of interruptions are non-toxic, enrol n/f patients to obtain n usable ones. At a plausible $f \approx 0.3\text{--}0.5$, that is roughly 15–35 patients enrolled to yield the 7–10 the design requires. Still a small study.

This tightens the design rather than weakening it, and it yields a new falsifiable prediction (§9): *within the same cohort, $\hat{c}$ estimated from toxic holds should be systematically lower than $\hat{c}$ estimated from non-toxic holds*. If the two strata agree, the confound is absent. If they diverge, the design has detected its own principal threat — from the inside, using data it was already collecting.

7 What actually drives adaptive therapy

7.1 Competition, not cost

Sweeping the fitness cost c (including **negative** values, where resistance is advantageous) against the competition coefficient α , restricted to the **stable-coexistence regime** ($\alpha < 1$ — above that the populations cannot coexist at all, which is the thing adaptive therapy depends on).

Absolute gain (days) of model-free adaptive therapy over MTD:

$\alpha \downarrow / c \rightarrow$	−0.20	0.00	+0.20	+0.40
0.10	−1	0	1	4
0.40	8	12	17	27
0.70	16	30	33	51
0.85	30	36	39	66

Leverage of α : **43 days**. Leverage of c : **22 days**. **Ratio $\approx 2 : 1$** .

1. **A fitness cost is not necessary.** 36 days of benefit at $c \leq 0$, where resistance is free or *advantageous*.
2. **A fitness cost is not sufficient.** At weak competition, a *large* cost ($c = +0.40$) buys **4 days**.
3. α carries roughly twice the leverage.

The field’s “if and only if” is false in both directions.

Sensitive cells suppress resistant ones primarily by **occupying space they need**, not by **outgrowing** them. A fitness cost amplifies the effect; it does not create it.

This is not without empirical precedent. Bacevic et al. [11] report that *spatial* competition alone constrains resistance to targeted therapy — a result our audit predicts, and which sits awkwardly with a literature that continues to frame adaptive therapy as contingent on a fitness cost.

7.2 This is not an artefact of Lotka–Volterra

In LV, α and c are independent free parameters — and that independence is what makes the comparison possible. So we rebuilt the test in **MacArthur’s consumer–resource model** [23],

the mechanistic model from which LV competition coefficients are *derived*, with $\alpha \propto$ niche overlap ω :

$$\text{uptake}_S = a_S (\text{Res}_1 + \omega \text{Res}_2) \quad (5)$$

$$\text{uptake}_R = a_R (\omega \text{Res}_1 + \text{Res}_2), \quad a_R = a_S(1 - c) \quad (6)$$

Here competition **emerges** from resource sharing rather than being assumed; $\omega = 0$ is a **genuine zero-competition control** (the clones eat different resources); and c is a separate trait (uptake efficiency), not collapsed into competitive ability.

ω	competition	$c = -0.10$	0.00	+0.30
0.00	NONE	−0	−0	−0
0.50	moderate	96	107	240
1.00	maximal	124	147	298

	Lotka–Volterra	MacArthur
ratio competition : cost	1.95	1.65
benefit with no competition , best c	≈ 0 d	exactly 0 d
benefit with no fitness cost	36 d	148 d

The result replicates. Different model, different mechanism, a real control — the same conclusion.

7.3 Scheduling needs no model

Adaptive therapy’s clinical rule is **closed-loop**: treat until burden falls to 50% of baseline, stop, resume at baseline — the rule used in the metastatic prostate cancer trial [4]. It has *no parameters*.

Truth	MTD	Gatenby (no model)	Open-loop (misspec.)	Oracle
continuum	637 d	847 d	651 d	882 d

The model-free feedback rule captures **86%** of the oracle’s available gain. The open-loop schedule derived from a misspecified model captures **6%** — while looking principled.

Feedback substitutes for identification — in *execution*, not in *selection*. You do not need an accurate model to *run* the therapy, given an accurate sensor and a stabilising feedback law. You still need α to know *whom to run it on*.

This distinction is easy to miss and it does not undercut the trial — it is the trial’s whole justification. A feedback controller is only worth deploying in a tumour where competitive suppression actually operates; at $\omega = 0$ or $\alpha \approx 0$ the loop captures nothing (§7.1–7.2), and the patient has been managed at sub-maximal dose for no benefit. **Identification is what tells you which cancers, and which patients, are candidates for the loop in the first place.** The measurement is for *selection*; the feedback is for *execution*. Neither replaces the other.

8 The proposed design

The audit demands a drug-off window, which looks like an intervention. It is not — because **the perturbation already exists**.

Treatment interruptions occur constantly in routine oncology for reasons that have nothing to do with evolutionary theory. From a system-identification standpoint these are **free excitations of the system**, and they are currently wasted, because nobody samples around them.

But — and this is the correction forced by §6.4 — **not all of them are clean**. The identifying set must be restricted to interruptions that perturb *the drug and nothing else*:

Clean (use these)	Confounded (exclude, or stratify separately)
insurance lapse / prior-auth delay	Grade 3/4 toxicity hold
drug supply interruption	hepatotoxicity, neutropenia, sepsis
scheduled surgery or elective procedure	severe fatigue / performance-status collapse
travel, clinic closure, holiday	intercurrent serious illness
non-adherence, physician discretion	dose reduction for tolerability

A hold caused by an insurance company tells you nothing about the patient’s physiology. A hold caused by Grade 4 neutropenia tells you a great deal, and all of it is confounded with what you are trying to measure.

We do not ask you to stop the drug. We ask that when you stop it — for whatever clinical reason you judge appropriate — we may draw blood densely around that window, and that the assay resolve clones rather than only total burden.

Concretely:

- (a) A tumour-informed panel reporting **each resistance variant separately** — not tumour-agnostic total ctDNA, and not a single merged allele fraction. Serial ctDNA tracking of specific resistance variants is already established practice [21, 22, 24]; what is missing is that the variants be reported separately rather than merged. This is the difference between recovering the clone structure and recovering nothing, at zero extra cost in phlebotomy.
- (b) ~18 draws per patient, concentrated in a **burst** bracketing any qualifying interruption, rather than spread uniformly.
- (c) **The window must be measured from drug clearance, not from the last dose.** $u(t)$ is not a step function: an agent with a multi-day half-life is still exerting selective pressure well after the pill stops. The effective drug-off window is the nominal hold *minus* the pharmacokinetic washout. Since 21 days is insufficient and 28 sufficient (§5.2), the operative requirement is **28 days of cleared drug** — for a long-half-life agent this may mean a nominal hold of 35–40 days, and for a very long-lived agent the design may simply not be feasible.
- (d) **Clone number fitted, not assumed**, with a continuum model in the candidate set; c_{eff} and α_{SR} as the estimands, not per-clone costs.
- (e) **No alteration whatsoever to the treatment decision.**

This preserves every property that made the observational framing worth having. No clinician is asked to accept any evolutionary hypothesis — one may believe adaptive therapy is nonsense

and still have no objection to the measurement; indeed, *if* one believes it is nonsense, this is the study that would show it. No therapeutic risk is introduced. It composes with existing trials as a companion sampling module. And it is falsifiable.

~89,000 patients under current practice versus ~7–10 under this design, for the same scientific question.

9 A falsifiable prediction the design tests on itself

§6.4 leaves the design exposed to a confound it cannot fully model. It can, however, **detect** it, using data it is already collecting:

Prediction. Within the same cohort, $\hat{c}_{\text{eff}}$ estimated from *toxic* holds will be systematically *lower* than $\hat{c}_{\text{eff}}$ estimated from *non-toxic* holds.

If the strata agree, the host confound is absent and the toxic holds can be pooled. If they diverge — in the predicted direction — the design has detected its own principal threat from the inside, and the toxic stratum is discarded.

This is not a patch. It is the strongest property the design acquired from being attacked: the confound that could invalidate it is *itself measurable* with the same eighteen blood draws, because the cohort naturally contains both hold types.

10 Limitations

- **Simulation is not evidence about patients.** Everything here is a statement about information in a model, not a finding about cancer. That is exactly the claim we intend, and no more.
- **This is a cohort design, not a personalised one.** No single patient’s data identifies α_{SR} .
- **The ontology is genuinely undecidable, and we route around it rather than solving it.** A reader who wants to know whether resistance in a given tumour is clonal will get no help here.
- **We did not vary the progression endpoint or the feedback rule.** All results use a burden threshold and the Gatenby-50% rule. These are the field’s conventions, not established facts.
- **c_{eff} is coarser than what the models want.** It answers whether the resistant mass grows more slowly — enough to decide whether adaptive therapy can *exist*. Fine-grained *scheduling* would want clonal detail we have shown to be unidentifiable; §7.3 suggests this may not matter, since feedback beats model-based scheduling anyway.
- **No immune compartment.** Both model classes are bipartite (sensitive vs. resistant); immune predation is absent. Note, however, that immune pressure acting on *total burden* is structurally a host effect of the kind analysed in §6.4: benign if it acts symmetrically on the clones, dangerous if it acts *differentially*. Since immune escape and drug resistance can co-occur, the differential case is plausible, and a three-compartment audit with an immune predator is the most important extension we have not done.
- **Shedding is assumed to scale linearly with burden.** Necrotic cores and vascularised margins plausibly shed at different rates, making β a function of tumour size and structure.

We tested misspecification of the shedding *ratio* (ρ) and the death-coupling (κ), but not *nonlinearity* in the burden–shedding map. This is untested and could matter.

- **The continuum rescue assumes resistance is smoothly distributed.** If resistance arises from a single catastrophic macro-evolutionary jump to a distant phenotype, fitting a continuum could smooth the likelihood surface and yield a *false positive* for identifiability. We tested discrete truths at $k = 2$ and $k = 3$; a single, extreme, isolated resistant phenotype is a case we did not test.
- **The models are well-mixed; real tumours are spatial.** α is a mean-field summary of what is physically a local, geometric process. Since Bacevic et al. [11] find that *spatial* structure constrains resistance, our well-mixed treatment probably *understates* competition — but “probably conservative” is not a substitute for having done the spatial version.
- **Parameter sweeps are evidence, not proof.** We varied c (0.05–0.40), assay CV (30–50%), shedding ratio (0.5–3), κ (2–20), clone number (2–3), competition (0–1.25), and model class (LV, MacArthur). The conclusions were stable across all of it.

PART II — HOW WE GOT HERE

11 Hypotheses, failures, and revisions

Part I is what survived. This is what did not.

We report it in full for three reasons: the failures are more instructive than the successes; a reader is entitled to know how many times the authors were wrong before arriving at a conclusion they are now asked to trust; and the *pattern* in the failures is this paper’s most transferable finding.

H1 — “Oncology samples too slowly. Denser sampling will close the gap.” — **REFUTED**

This was the paper’s founding thesis. It was wrong.

The reasoning was seductive: ctDNA clears from plasma with a half-life of minutes to hours, the dynamics of interest run over days, and restaging happens every eight weeks. A high-bandwidth sensor read at low bandwidth by convention. Fix the cadence, fix the problem.

What broke it. We tested it. Six draws $\rightarrow$ 281 draws (daily, for forty weeks). The sign-error rate went from $\sim 57\%$ to $\sim 57\%$. *Exactly unchanged.*

What replaced it. Sampling density is *necessary but nowhere near sufficient*. The binding constraint was never cadence — it was that we were observing an **unperturbed** system through an observable that **cannot see the clones**. Hence the triad (§5). Density is one leg of three, and worthless on its own.

Why we got it wrong. We reasoned from an analogy — signal processing, Nyquist — to a system that was not bandwidth-limited but *information*-limited. The analogy was elegant and irrelevant.

H2 — “Clone number is recoverable with the right panel.” — **REFUTED**

Having found that a fixed compartment count destroys the estimate, we proposed inferring it. We showed a per-variant panel recovers the true clone number 100% of the time where a lumped allele fraction recovers it 0%, and declared the hazard closed.

What broke it. The candidate set. We had fitted $k = 1, 2, 3, 4$ — all discrete. Adding a **continuum** model, BIC prefers it in *both* worlds, including 6/8 against a genuinely three-clone truth. We had been asking *which* discrete model while assuming the answer to *whether*.

What replaced it. The ontology is undecidable, and does not need to be decided (§6.3). Re-specify the estimand as a *functional* of the resistance distribution and the sign comes back correct under either ontology.

Why we got it wrong. We let the candidate set encode the hypothesis. A model-selection procedure can only tell you which of the models you *offered* it fits best. Offering only discrete models and concluding “the tumour is discrete” is circular — and we did it without noticing.

H3 — “The cost of resistance is the quantity to measure.” — REFUTED

This was not merely our assumption; it is the field’s. Adaptive therapy is standardly justified as working *if and only if* resistance carries a fitness cost.

What broke it. In a world where resistance was entirely *free* ($c = 0$), the model-free adaptive rule still beat MTD substantially. If it works when resistance costs nothing, the cost is not the mechanism.

What replaced it. α (§7). Sensitive cells suppress resistant ones by **occupying carrying capacity**, not by outgrowing them. The cost amplifies this but does not create it: it is neither necessary (36 days of benefit at $c \leq 0$) nor sufficient (4 days at large cost, weak competition), and competition carries $\sim 2\times$ the leverage. Replicated in a mechanistic consumer–resource model with a genuine no-competition control.

Why we got it wrong. We inherited the field’s framing without auditing it. The identifiability machinery we had built to *test* the field’s parameter is what eventually revealed the parameter was the wrong one — but only because we ran a sweep we had no particular reason to run.

H4 — “The benefit is flat in the cost of resistance.” — REFUTED (our own arithmetic)

Having discovered H3, we overshot. We reported the benefit as a **relative** gain, $(TTP_{\text{adaptive}} - TTP_{\text{MTD}})/TTP_{\text{MTD}}$, observed it was flat across c , and concluded the fitness cost was simply irrelevant.

What broke it. The denominator. MTD survival *also* rises with c (resistant cells grow more slowly, so even MTD does better), masking a real effect in the numerator. In **absolute days**, the benefit roughly **doubles** across the range of c .

What replaced it. The corrected, weaker, true claim: c is not *required*, but it is not irrelevant either. It roughly doubles the benefit across its range; α carries twice its leverage; neither is negligible.

Why we got it wrong. We chose a normalisation that flattered a conclusion we had just become excited about, and did not check whether the denominator was moving. This is the most ordinary statistical error in this document, and it produced the most confident-sounding sentence in the draft it appeared in.

H5 — “ α_{SR} is structurally unidentifiable.” — REFUTED (our own arithmetic)

While auditing H4 we discovered our α results assumed *symmetric* competition. Relaxing that, the directional coefficient α_{SR} appeared unrecoverable: $SE \approx 1.5$, interval spanning zero *and negative* — the data apparently could not distinguish suppression from assistance. Extra drug-off windows did not help, suggesting the barrier was structural. We wrote it up as the paper’s most serious limitation.

What broke it. Two things, both embarrassing.

1. Those intervals were **per-patient**. Every other parameter in this paper is unresolved per-patient and becomes resolvable by pooling — c needs $n \approx 7$. We had silently demanded that α_{SR} clear a bar in one patient that c had never cleared, then called the failure structural. Pooled across $n = 10$, α_{SR} **resolves cleanly**: $[0.472, 1.941]$.

2. The SEs came from pseudo-inverting a Fisher matrix with condition number $\sim 10^7$ — and they were **non-monotone in the amount of data** ($1.507 \rightarrow 1.418 \rightarrow 1.553$ as holds went $1 \rightarrow 2 \rightarrow 3$). *More data cannot reduce information.* That non-monotonicity was the numerics telling us the bound was garbage. We read it as a finding.

Why we got it wrong. We committed, twenty minutes later, exactly the error our own §4.2 warns against — trusting a Cramér–Rao bound on a near-singular surface. Writing the warning did not immunise us against the mistake.

A negative result worth keeping. We hypothesised that reparameterising to $\sigma := r_R \alpha_{SR}$ — the directly observable slope $-\partial g_R / \partial S$ — would be identifiable where α_{SR} was not, since the data only ever see the product. **It buys nothing.** σ needs $n \approx 11$, essentially identical. Because r_R is adequately determined under this design, σ and α_{SR} carry the same information.

H6 — “Vary the resource supply rate to vary competition strength.” — MIS-SPECIFIED

Testing whether α -dominance survived outside Lotka–Volterra, our first mechanistic model varied competition by varying the resource **supply rate** ϕ .

What broke it. The R^* rule [25]. In a consumer–resource system the equilibrium resource level is $\text{Res}^* = Km/(r - m)$, which is **independent of supply**. Raising ϕ raises equilibrium *biomass*, not resource *availability*. Every arm remained fully resource-limited; there was never a no-competition control.

The mis-specified experiment reported an apparent $11\times$ **inversion** of the α result — a clean-looking, entirely spurious finding.

What replaced it. Niche overlap (§7.2), which *is* the mechanistic origin of α , and which gives a genuine $\omega = 0$ control. The result then replicated.

Why it matters. This error produced a confident, publishable-looking table pointing the opposite direction from the truth. Nothing in the output looked wrong. We caught it only by asking *what does this knob actually do to the equilibrium?* — a question we should have asked before running the sweep, not after.

H7 — “Any incidental treatment interruption is a free excitation.” — REFUTED (by a referee, correctly)

Every draft through v1.0 listed toxicity holds first among the interruptions the design would exploit. A referee pointed out that this is biologically naive: a Grade 4 neutropenia hold means the patient is in systemic crisis, and the growth and competition parameters of a tumour in a metabolically deranged host are not those of a tumour in a well one. We had been treating a physiological catastrophe as a clean laboratory perturbation.

What broke it. We modelled it (§6.4). The critique is correct, and the damage is severe — but the *structure* of the damage is sharper than the critique stated. A **symmetric** host insult is comparatively benign, because c is a *ratio* and a common factor cancels. A **differential** insult that spares the resistant clone is fatal: sign-error goes to **58%**, and $\hat{c}$ is biased **downward** — i.e. toward concluding resistance is free, which is exactly the error the paper exists to prevent. Fitting the host effect as a nuisance parameter only partly helps ($36\% \rightarrow 22\%$).

What replaced it. An inclusion criterion, not a statistical fix. Only interruptions that perturb *the drug and nothing else* — insurance lapse, supply interruption, scheduled surgery, travel,

non-adherence — are clean excitations. Toxicity holds are excluded from the identifying set. Enrolment rises by $1/f$; validity is preserved. And the design gains a way to detect the confound on itself (§9).

Why we got it wrong. We were reasoning about $u(t)$ and forgot there was a patient attached to it. The model had a drug term and a tumour term and no host, so the host could not appear in the analysis — a category of error no amount of internal verification would ever have caught, because the machinery was working perfectly on the wrong system. It took someone outside the model to see it.

11.1 What survived, and the pattern

Every substantive thesis in this paper’s first five drafts was refuted. The design recommendation survived all of them — and then a referee found the one thing that could break it.

That asymmetry is worth dwelling on. The *estimand* moved three times (from c , to c_{eff} , to α_{SR}). The *ontology* collapsed. The *mechanism* inverted. But “clone-resolved observation + a drug-off window + burst sampling around it, at ~ 18 draws and ~ 10 patients, purely observationally” was right from the first audit and never moved.

H7 is the exception, and it is the instructive one. It did not come from us. Six rounds of adversarial self-audit, a verification suite that checks the Fisher information against the numerical Hessian to 2×10^{-8} , and three independent implementations of the model agreeing to 10^{-6} — none of it could find H7, because none of it was looking outside the model. The machinery was working perfectly on a system that was missing a host. **Internal rigour cannot find a missing compartment.** Only a reader who knows what a Grade 4 neutropenia hold actually does to a patient can find that, and no amount of Monte Carlo substitutes for one.

This is what one would hope for from an identifiability analysis. **Conclusions about what a protocol can see are far more robust than conclusions about what the world is.** That is an argument for doing the audit first.

And the pattern in the failures:

Failure	Category
H1 — sampling density	Reasoned from a wrong analogy (bandwidth, not information)
H2 — clone number	Let the candidate set encode the hypothesis
H3 — cost of resistance	Inherited the field’s framing without auditing it
H4 — “flat in c ”	Chose a normalisation that flattered the conclusion
H5 — α_{SR}	Trusted a bound we had ourselves warned against
H6 — supply rate	Did not ask what the knob controlled
H7 — toxicity holds	Modelled the drug and the tumour, and forgot the host

Not one of these was a coding error — and the most dangerous one was not even a reasoning error *within* the model. It was a missing piece of biology.

We ran an independent verification suite over the machinery: three model implementations cross-checked to $< 10^{-6}$; the piecewise integrator against brute-force Euler to 1.3×10^{-4} ; the Fisher information against the numerical Hessian of the expected likelihood to 2×10^{-8} ; the allele-fraction observable checked for saturation; the headline result checked across independent seed sets. **All passed.** The code was right every time.

Every error was one of interpretation, or of omission. That is almost certainly where the errors in this literature live too — and it is the strongest argument we can make for why an identifiability audit of *other people's* models is worth doing at all. It is also an argument for showing your audit to someone who knows the biology, and believing them when they tell you the perturbation you have been counting on is a sick patient.

12 Conclusion

Oncology cannot currently measure the parameters its evolutionary models depend on. Not the cost of resistance — roughly half of honest fits get its sign wrong. Not the competition coefficient — its estimate wanders the entire admissible range. And not at any sampling rate, because the system is never perturbed and the observable cannot see the clones.

The repair is cheap and requires no clinician to accept anything. Resolve the clones. Sample densely around the drug-free windows that routine care already hands over for nothing. Ten patients, eighteen draws each, no change to any treatment decision.

And the quantity the field has organised itself around — the fitness cost of resistance — is not the one that decides the question. Competition is. A fitness cost is neither necessary nor sufficient for adaptive therapy to work; it merely amplifies an effect that comes from sensitive cells occupying space that resistant cells need. This holds in the field's own model class, and replicates in a mechanistic one with a genuine control.

We would not want a reader's takeaway to be any single number in this paper; §11 records how many of them we have had to withdraw. The takeaway is the method:

**Ask what your measurement protocol can identify,
before you argue about what your model implies.**

A decade of argument has been conducted over parameters that, under the field's own model and its own data, could not have been measured either way.

Code, models, and the full audit — including every experiment that refuted an earlier draft — are reproducible from the accompanying scripts. Correspondence: ORCID 0009-0006-6283-1696.

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